

Palm OS® Cobalt Technical Introduction

… and conference roadmap

Presented By:

David Fedor
Director, Developer Architecture and Disclosure

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Tagline for the Developers Conference?

"All the Palm OS® Cobalt information that your brain can hold"



The Big Picture

- **10+ hours devoted to Palm OS® Cobalt presentations**
  - Enough to know the new capabilities available to you
  - Enough to get you addressing any compatibility issues
  - Jumpstart on new features
- **Tools, simulators, hardware, engineers…**
- **Devices coming later this year**



The Big Picture (cont.)

- **Palm OS® Cobalt includes an extensive infrastructure update**
  - Hundreds of new APIs available
  - Designed for strong compatibility
- **ARM-only**
- **Might run on existing ARM-based devices**
  - Depends on licensee choices
- **New Tools**
  - IDE, compilers, linkers, etc.
  - Resource tools
  - Simulator



Change is threatening

People want big improvements.

People want nothing to change.

So… nothing is changing,
and everything is changing.



Changing everything, but nothing

For PACE applications, the world doesn't change

- (much.)
- This is a perfectly valid model to use for years.

New APIs and capabilities are available to Palm OS® Protein applications

- This does involve writing new code and tweaking old code.
- "Protein" indicates the fully native ARM applications using the new capabilities introduced in Palm OS® Cobalt version 6.0



"Sneak Peek" Features

Protected memory system

Process protection

- (stability and security)

New runtime model for flexible apps & libraries

Extended database architecture

- (interoperability and extensibility)

Threading and Background processes

New Communications architecture

… and many more



More than the Sneak Peek

- Status Bar and Slips
- Scalable Fonts
- GC APIs
- Automatic back-buffering
- DIA and Pinlets
- Multiprocess UI and layers
- HotSync Exchange
- Multimedia
- IPC, keys, memory architecture
- Trusted desktop
- Mem/Dm split, legacy heap
- Db details
- Threading details, event queues, keepers
- IOS, new CM UI
- AM, AZM
- Remote sublaunching



Existing Application Compatibility

Compatibility is similar to Palm OS 5.x situation

- PACE runs your existing 68k PRCs
- PNOs in your 68k PRCs still ideal for speed optimization

Assumptions about OS implementation are dangerous

- The internals changed a lot
- Deprecated and unsupported APIs were removed
- Public APIs are supported

High degrees of compatibility will be expected

- (Assuming your applications are well behaved!)



API Compatibility

Basic strategy: compatibility with 5.x APIs

A few APIs were unsupportable

- Blatant security holes
- Some APIs assumed the old memory model

Fix several problems which developers have complained about for years

Large majority of old APIs are identical

- You'll hear nothing but changes and new APIs today, but that doesn't mean everything changed!

PACE still is there



Compatibility

Test for compatibility ASAP

- Use the simulators
  - Also double-check on a new Palm OS® Garnet simulator
- Use the hardware in our lab tonight
- Use the self-test
- Use the documentation
  - Particularly "Porting Applications to Palm OS Cobalt"



New Tools on the Conference CD

- **Palm OS® Cobalt Simulator**
- **Palm OS® Developer Suite**
  - Palm OS® Eclipse
  - Palm OS® Protein C++ Compiler and tools
  - Palm OS® Debugger
  - Palm OS® Resource Editor and tools
  - Palm OS® Protein SDK
  - Documentation
- **CDK**
- **Palm OS® Garnet simulator**
- **Samples, FAQs…**
- **See presentation 305 later today**



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Demo #1 !



Switching to Palm OS® Protein

Why do it?

- To use new features
- Increase speed
- Removal of constraints, cleaner code

Why not do it?

- A single 68k PRC can be everywhere (and can include a PNO)
- If you (or your customers) are conservative



Protein empowerment

A few standard steps to get to Palm OS® Protein

- Mechanical changes (search and replace)
- Stop using the deprecated (or unsupported) functions
- Address endianness and structure packing in your saved data
- Make changes to work best with the new environment, e.g.:
  - Specify Database Reference along with resource IDs
  - Use frmUpdateEvent instead of save-behind or WinSaveBits()
  - Specify which special launch codes you care about
  - Use Berkeley sockets instead of NetLib

See presentation 300 tomorrow, and the porting guide



Palm OS® Cobalt Processes and Memory Mapping

Data Mgr	System	I/O	Application	Background	Sublaunch x3	Licensee
Data Mgr threads	Window Mgr	IOS	UI application	Comms thread(s)	Notification Agent	License UI Component
	App Mgr	VFS	PACE	Exchange Sublaunch	Find Agent	Licensee Srv
	Event Mgr	STREAMS framework	OS Shared libraries	Slips	PACE	
	AM/AZM	STREAMS components		Sound callbacks		
	Status Bar	HW device drivers		PACE		
	Feature Srv					
	Exchange Srv					
	Notification Srv					
Local Heap	**Local Heap**	**Local Heap**	**Local Heap**	**Local Heap**	**Local Heap**	**Local Heap**

System Shared Memory

Secure DBs | **Frame Buffer** | **Audio Mixer** | **Legacy Heap (on request)**

Storage Heap

ROM



Core OS

Multiple processes, protected from each other

- Designed to minimize code running in “privileged” mode
- Kernel-managed, consistent access control to protected resources

Palm OS APIs are all accessed through shared libraries

- Some APIs execute within the app process
- Some are a gateway to another system process (IPC)

Threading

- Each thread gets its own events

New memory management system

- Heaps up to 256 MB, allocations up to 16 MB

See presentation 310 later today



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Demo #2 !



New Drawing Framework

GC: “Graphics Context”

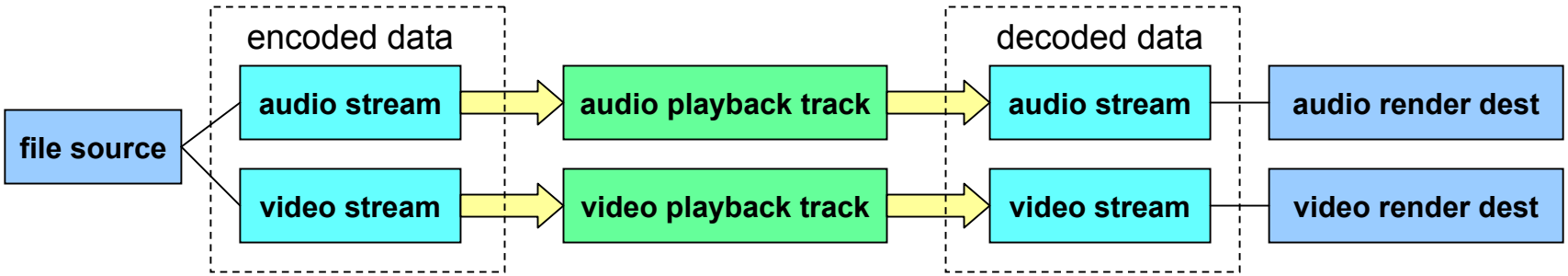
- New 2d drawing model
- TrueType fonts, path filling, rotation, anti-aliasing…
- Very similar to OpenGL
- The old Window Manager APIs are now built upon it

```
gc = GcGetCurrentContext();
GcSetFont(gc, theFont);
GcDrawTextAt(gc, x, y, theStr, strlen(theStr));
GcStroke(gc);
GcReleaseContext(gc);
```

Multimedia framework

Control video and audio media tasks

- Flexible system of sources, destinations, and tracks



See presentation 315 tomorrow



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Demo #3 !



New Window Manager

Biggest change: the UI is now multithreaded

- Doesn't all depend on application's event loop
- Clipping enforced to your drawing window

New window layout

- Active Input Area
- Status Bar
- Slip windows
- Independent layers

Constraints-based layout

- Allows dynamic behavior based on user action



New Window Mgr (cont.)

320x320 and 320x480 common

- Other modes might be available, but less expected
- 16-bit mode might well be always active
  - (you can ask to switch to a lower mode, but it won't do anything)

Some windows can be back-buffered

- Mostly for compatibility

Update-based windows are better

- Owner must always be able to draw when asked
- Usually can't be read from
- Much better for hardware acceleration



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Demo #4 !



Communications architecture

All new architecture based on open standards

- Berkeley sockets, POSIX calls, standard I/O (libc: fprintf())…
- Fits well with new threading capabilities
  - Have one thread blocked waiting for input; another doing UI.
  - Alternatively, poll using PollBoxes or sometimes IOSPoll()
  - Run inside your application's process, or in background
- Compatibility libraries exist for most systems, except NetLib



Communications (cont.)

Connection Manager (and application)

- Consolidates the UI from the Connection, Bluetooth, Network, and Phone Prefs panels
- Profile example: "this serial port, 9600 baud, even parity" or "TCP/IP connection over Bluetooth to this access point"
- Based on a flexible framework
- Apps can create, establish and manipulate connections, & override UI

See presentation 320 on Thursday



Three Types of databases

Classic: same as 5.x and prior

- Records and resources

Extended: like Classic databases, but better

- Allows >64k records or resources
- Uniquely identified by db name PLUS creator ID
- Likely to have data in little-endian format

Schema

- The new kind! Flexible tables of data, extensible…



Schema Databases defined

A schema database contains tables (often just 1)

Tables consist of rows and columns

A column is a particular data type

- FirstName… birthdate… priority… note…

A row is like a record

The collection of all column descriptions for a table is called a schema

Extensible (and opaque) data format



Table example

Name	Age	Phone #
Joe	34	555-4857
Mary		(512) 555-8732
John	56	555.0254
Bob	19	800-555-4BOB
Ann	24	



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Demo #5 !



Schema Databases (cont.)

- **Automatic sorting with multiple indices**
- **SQL-like iteration and queries**

  SELECT * FROM mytable WHERE age < 21 ORDER BY LName
- **Built-in PIM applications use Schema databases**
  - So you can now safely add data to the PIM databases
- **The Db*() APIs on device and in the CDK match**

See presentations 325 and 340 on Thursday



PIM databases and 68k apps

Applications which access PIM databases don't know about schema

- Would thus have compatibility problems

PACE to the rescue!

See presentation 330 tomorrow



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Demo #6 !



HotSync® Exchange

New transport in "Send…" menu, as well as extension of Desktop Install Tool

Allows data import/export without writing a conduit

See presentation 325 on Thursday



Security on Palm OS® Cobalt

Authentication Manager (“AM”)

Authorization Manager (“AZM”)

Database access rules

Backing up secure data

See presentation 335 on Thursday



Other topics you should know

Various other improvements for you

- Standard data types (stdint.h style)
  - uint8_t, int32_t and so forth
- libc
- STLPort

Performance differences for some APIs

- See presentation 340 on Thursday

PIM and Desktop changes

- See presentation 325 on Thursday



Next Steps

Get a CD now

- Join the seeding program for Palm OS® Garnet updates
- The Palm OS Developer Suite has built-in online updating

Come to the labs tonight and tomorrow night

- Test on boards and with your simulator
- Whip out some new features
- Engineers available to help & instruct
- Contest in the lab! Fabulous prizes!

Protein-forum@news.palmos.com next week!